

# ACTIVITY-5

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**COURSE CODE-** DSA 0603

**COURSE NAME-** DATA HANDLING AND  
VISUALISATIONS

## Activity 1: Geospatial Data Visualization

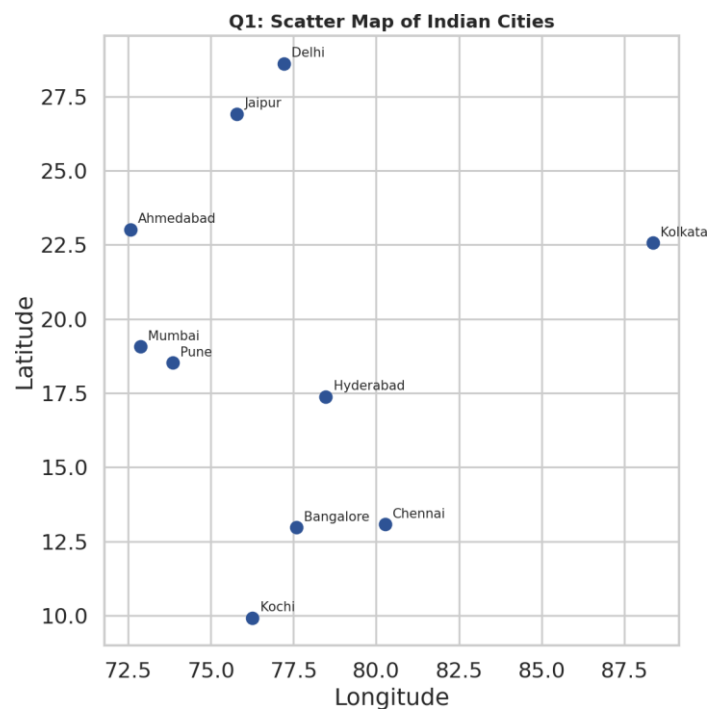


Figure 1.1 (Q1) - Scatter map of Indian cities by longitude/latitude.

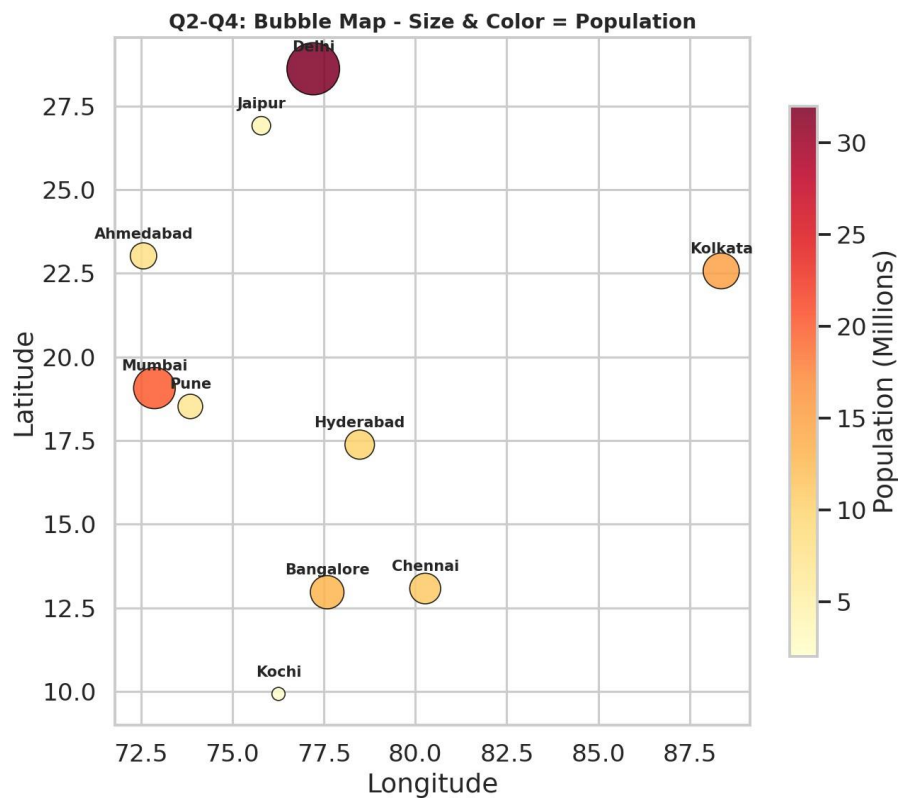


Figure 1.2 (Q2-Q4) - Bubble map: size & color = population, labeled with `geom_text()`.

**Q1.** Create a scatter map showing all cities.

See Figure 1.1 - each point is one city plotted at its (Longitude, Latitude).

**Q2.** Bubble map where bubble size represents population.

See Figure 1.2 - bubble radius scales with population (`geom_point(size = Population)`).

**Q3.** Label each city using `geom_text()`.

City names are overlaid above each bubble in Figure 1.2 via `geom_text(aes(label = City))`.

**Q4.** Different colors based on population size.

Bubble fill color follows a yellow-to-red gradient scaled to population (`scale_color_gradient()`) in Figure 1.2.

**Q5.** City with the highest population.

**Delhi**, with 32 million people - visibly the largest, darkest-red bubble in Figure 1.2, well ahead of Mumbai (20M) and Kolkata (15M).

## Activity 2: Network Visualization

Q1-Q2: Network Graph (highest-degree nodes in red)

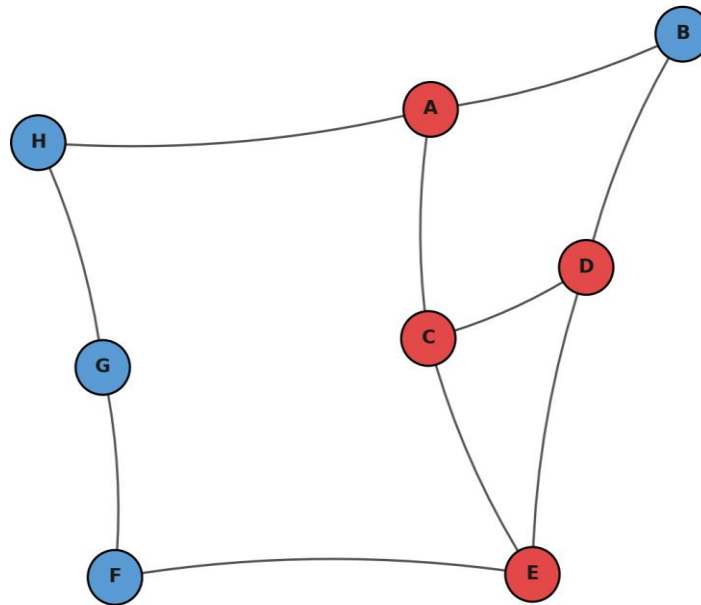


Figure 2.1 (Q1-Q2) - Directed network graph; highest-degree nodes highlighted in red.

**Q1.** Create a network graph using `igraph`.

See Figure 2.1 - built with `graph_from_data_frame()` and `plot.igraph()`.

**Q2.** Node labels and custom colors.

Node names are shown via `vertex.label`, and nodes are colored with `vertex.color` (red = highest degree, blue = others) in Figure 2.1.

**Q3.** Degree of each node.

Using `degree(g, mode='total')` (in-degree + out-degree): A=3, B=2, C=3, D=3, E=3, F=2, G=2, H=2.

(In-degree: A=1, B=1, C=1, D=2, E=2, F=1, G=1, H=1 | Out-degree: A=2, B=1, C=2, D=1, E=1, F=1, G=1, H=1.)

**Q4.** Node with the highest number of connections.

Four nodes are tied for the highest total degree of 3: **A, C, D, and E** - each is the endpoint of 3 edges combined (incoming + outgoing). This is visible in Figure 2.1 as the four red-highlighted nodes, which sit at the busiest junctions of the cycle-like graph structure.

**Q5.** Plot the network using a circular layout.

See below - built with `layout_in_circle(g)`.

**Q5: Network Graph - Circular Layout**

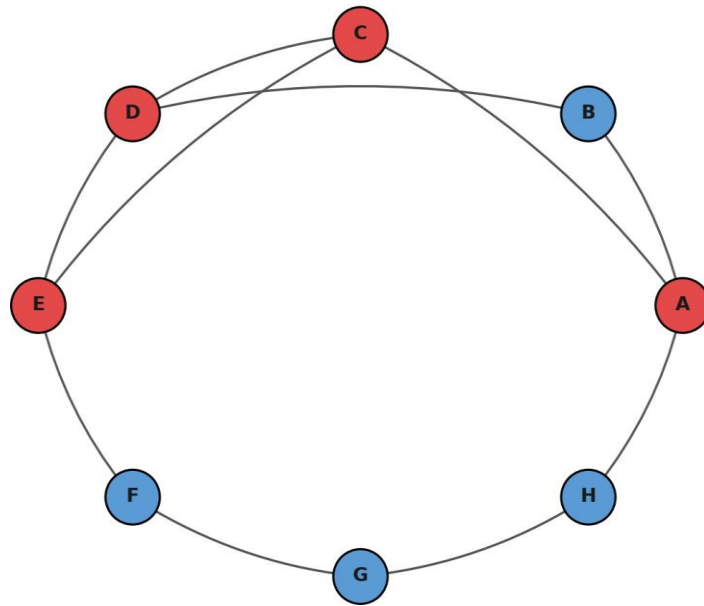


Figure 2.2 (Q5) - Same network drawn with a circular layout.

## Activity 3: Temporal Data Representation

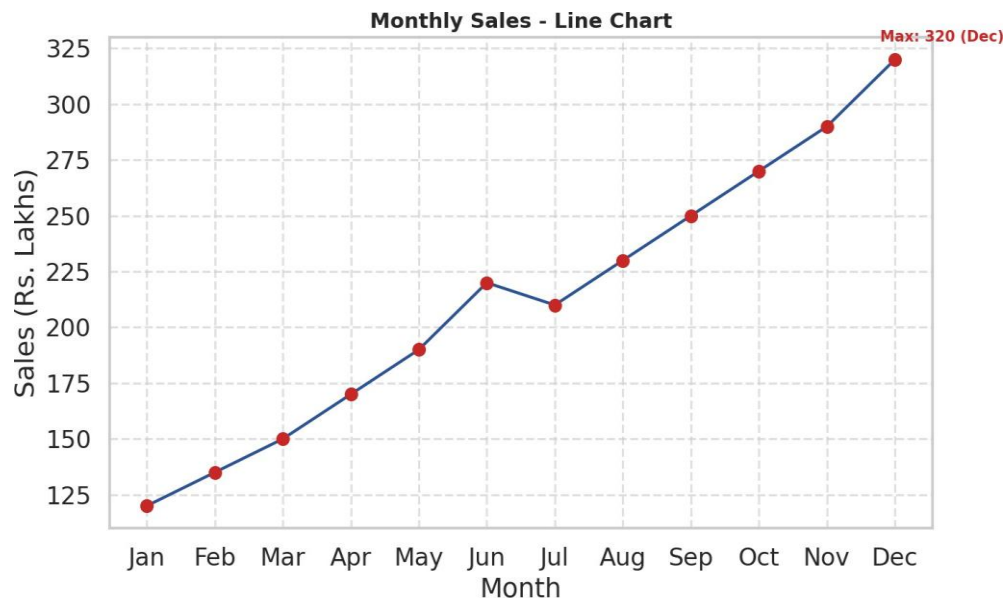


Figure 3.1 - Monthly sales line chart with grid lines, month labels, and highlighted points.

**Q1.** Line chart of monthly sales.

See Figure 3.1 - built with `plot(type='n') + lines()` to draw the connecting trend line.

**Q2.** Custom month names via `axis()`.

The x-axis uses `axis(1, at = 1:12, labels = months)` to show Jan-Dec instead of the numbers 1-12.

**Q3.** Highlight data points using `pch=19`.

Each month's value is marked with a solid filled circle (`points(..., pch = 19)`), shown in red in Figure 3.1.

**Q4.** Grid lines and proper axis labels.

Dashed grid lines are added with `grid()`, and both axes are labeled ("Month", "Sales (Rs. Lakhs)").

**Q5.** Month with maximum sales.

**December**, with Rs. 320 Lakhs - the peak of the whole year and the final point on the rising trend line in Figure 3.1, more than 2.6x January's Rs. 120 Lakhs.

## Activity 4: Multivariate Data Visualization

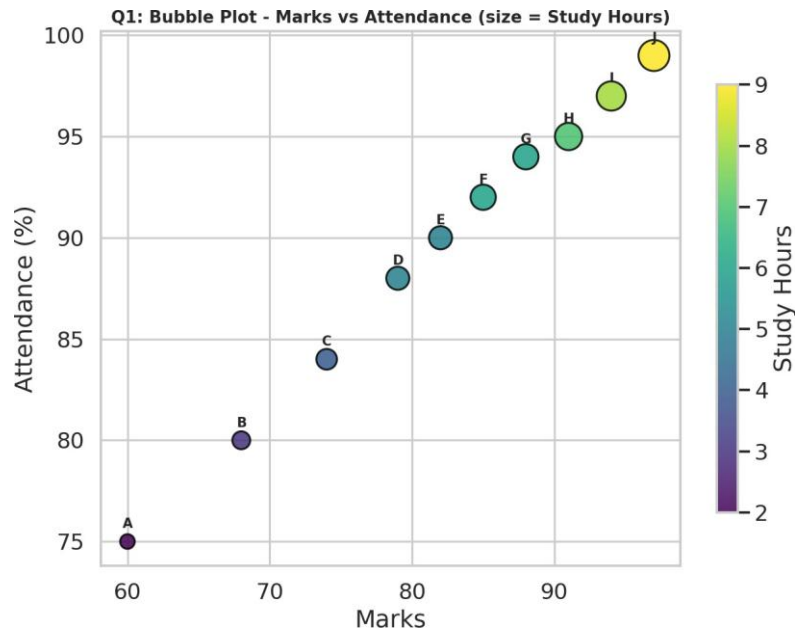


Figure 4.1 (Q1) - Bubble plot: Marks vs Attendance, bubble size = Study Hours.

**Q1.** Bubble plot - Marks (x), Attendance (y), size = Study Hours.

See Figure 4.1 - built with `geom_point(aes(size = StudyHours, color = StudyHours))`.

**Q2.** Scatter plot matrix using `pairs()`.

See Figure 4.2 below - every pairwise combination of Marks, Attendance, Study Hours, and Projects is plotted.

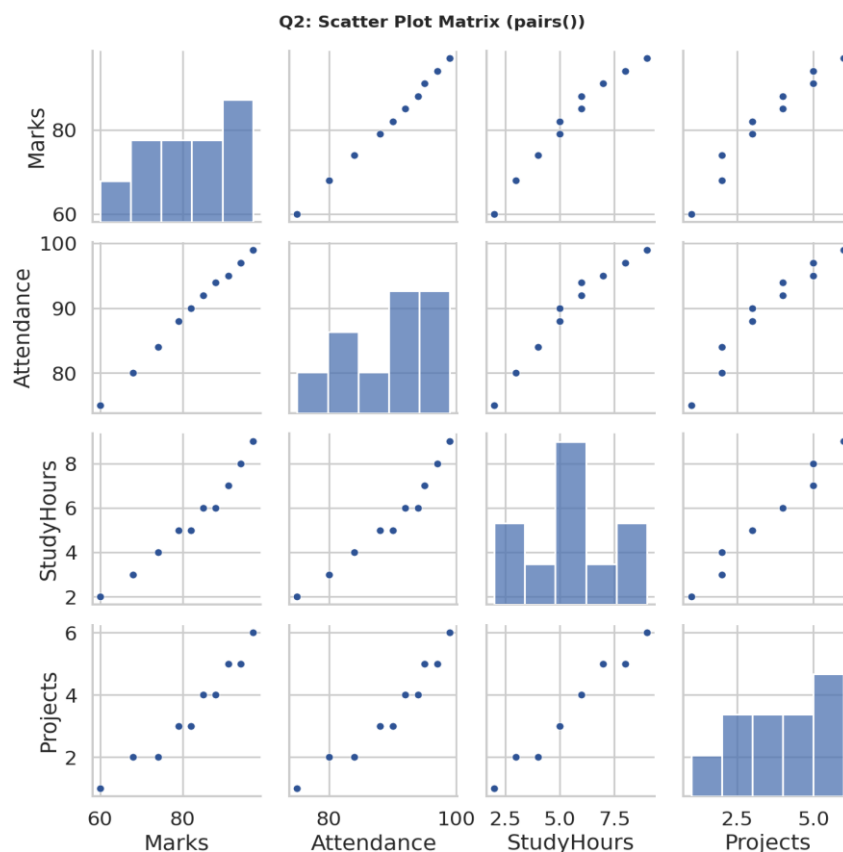


Figure 4.2 (Q2) - Scatter plot matrix (`pairs()`) of all four numeric variables.

**Q3.** Correlation among Marks, Attendance, and Study Hours.

`cor()` gives: Marks-Attendance = **0.998**, Marks-StudyHours = **0.981**, Attendance-StudyHours = **0.974** - all extremely strong positive correlations, meaning students who study more also attend more and score higher, essentially in lock-step.

**Q4.** Student with highest marks and largest bubble.

**Student J** - 97 marks (highest) and 9 study hours (also the largest bubble in Figure 4.1), confirming that the top scorer is also the one who studied the most.

**Q5.** Heatmap of relationships among all numerical variables.

See Figure 4.3 below - correlation heatmap across Marks, Attendance, Study Hours, and Projects.

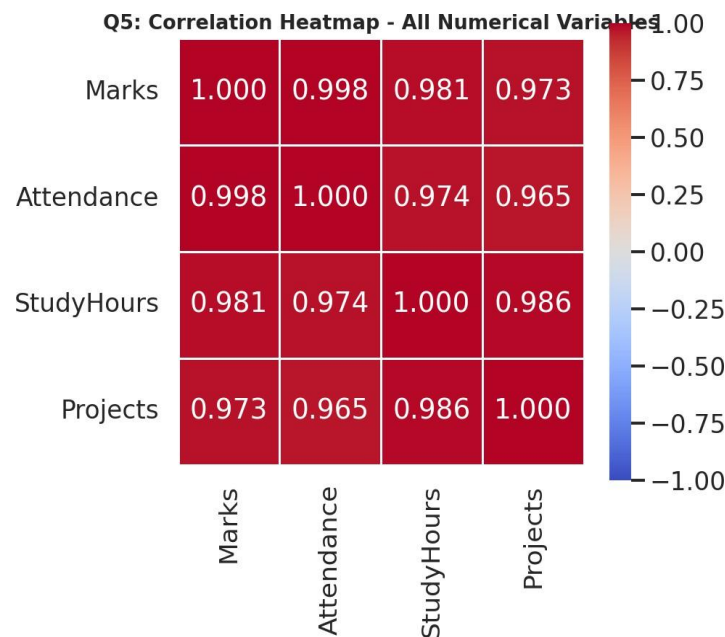


Figure 4.3 (Q5) - Correlation heatmap of all numeric variables (all pairs strongly positive,  $r > 0.96$ ).



## Activity 5: Integrated Data Visualization Exercise



Figure 5.1 (Q1) - Yearly sales line chart, 2016-2025.

**Q1.** Temporal line chart of yearly sales.

See Figure 5.1 - sales grow every year without exception, from Rs. 150 Lakhs (2016) to Rs. 480 Lakhs (2025), a 3.2x increase over the decade.

**Q2.** Multivariate bubble plot - Sales (x), Profit (y), size = Customers.

See Figure 5.2 below - bubble size scales with the Customers column, color encodes Year.

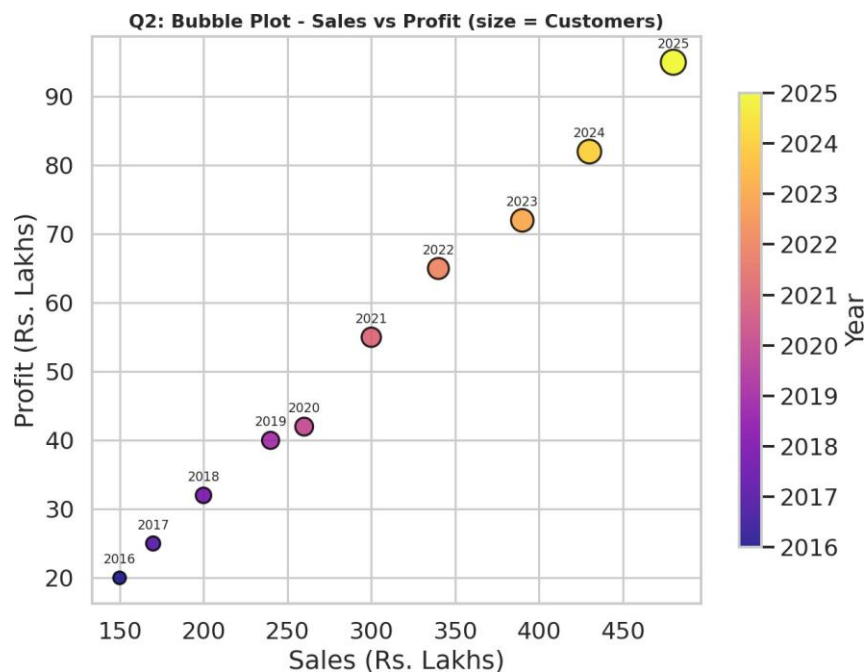


Figure 5.2 (Q2) - Bubble plot: Sales vs Profit, bubble size = Customers, colored by year.

**Q3.** Correlation matrix for Sales, Profit, Customers, and Branches.

cor() on all four variables gives: Sales-Customers = **0.9996**, Sales-Profit = **0.9982**, Profit-Branches = **0.9986**, Customers-Branches = **0.9959**, Sales-Branches = **0.9957**, Profit-Customers = **0.9981** - every pair is almost perfectly positively correlated, showing the company has scaled sales, profit, customer base, and branch count together in a highly consistent growth pattern.

**Q4.** Scatter plot matrix using pairs().

See Figure 5.3 below - every pairwise relationship shows a clean, near-linear upward trend, consistent with the near-1.0 correlations above.

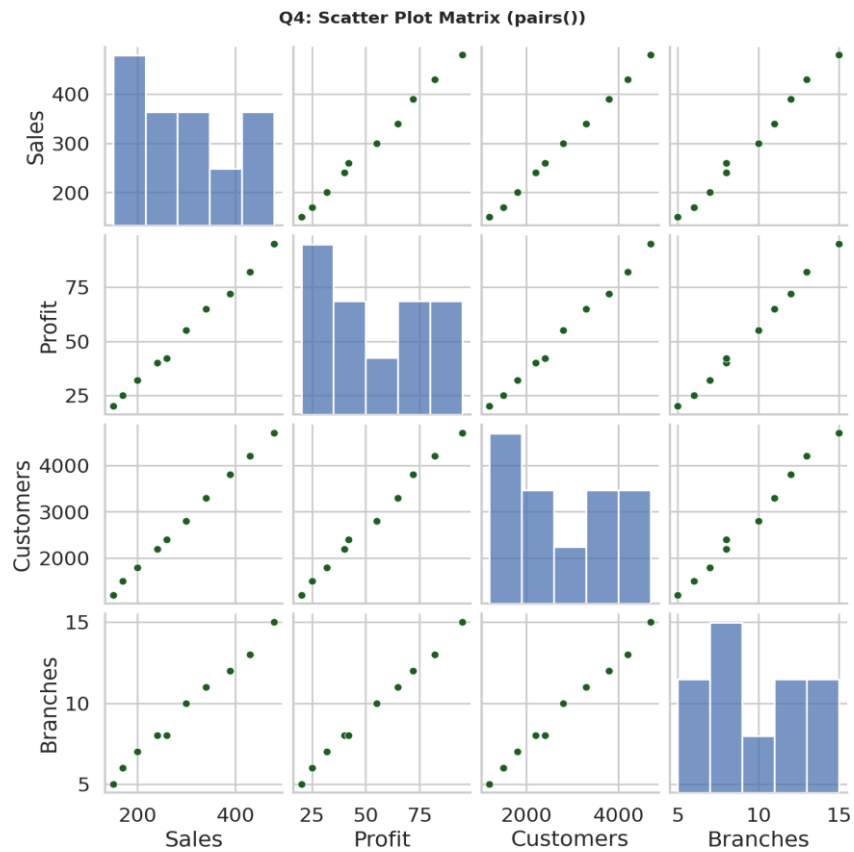


Figure 5.3 (Q4) - Scatter plot matrix (pairs()) of Sales, Profit, Customers, and Branches.